

BARSAT KHADKA

CONTACT

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EDUCATION

The University of Southern Mississippi 2024 – PRESENT
Hattiesburg, MS
B.S., Computer Engineering GPA: 3.92
THESIS: *Reinforcement Learning in World Models* (in progress)

FELLOWSHIPS & PROGRAMS

Break Through Tech AI/ML 2026 - 2027
Selected participant, [Cornell Tech] — applied machine learning fellowship.
SanDiego Supercomputer Center CIML JUNE 2026
SDSC CIML Summer Institute, UC San Diego — selected for funded HPC/ML training institute.
UR2PHD - Computing Research Association JULY 2026
Undergrad Research to Phd Program.

PUBLICATIONS

CTS-Bench: Benchmarking Graph Coarsening Trade-offs for GNNs in Clock Tree Synthesis
Khadka, B., Roxy, K., Ahmed, M.R.
ACCEPTED, MLBench Workshop @ ASPLOS 2026 arxiv.org/pdf/2602.19330

PREPRINTS & UNDER REVIEW

SwiftCTS: Fast Cross-Design Prediction and Pareto Optimization of Clock Tree Metrics via Few-Shot Calibration
Khadka, B., Ahmed, M.R.
UNDER REVIEW, ICCAD 2026 arxiv.org/pdf/2606.11348
SEU: Siloed Exact Machine Unlearning for Fine-Tuned Models
Maharjan, N., Khadka, B., Abdelfattah, R.
UNDER REVIEW (second author)
Filter then Verify: A Multiphase Social Engineering Attack Detection Framework using GNN and BERT for email networks
Khadka, B., Koirala, P., Neupane, K., Rahimi, N.
UNDER REVIEW, Springer arxiv.org/pdf/2605.17201
MechRL: Reinforcement Learning Agents Perform Circuit Discovery for Mechanistic Interpretability
Khadka, B. (single author)
UNDER REVIEW, TMLR arxiv.org/pdf/2605.26343

RESEARCH EXPERIENCE

Independent Research Collaboration with Dr. Rubel Ahmed 2025 – PRESENT
Louisiana Tech University (Remote)

- Co-developed CTS-Bench, a benchmark suite for evaluating graph coarsening trade-offs in graph neural networks applied to VLSI clock tree synthesis.

- Designed SwiftCTS, a physics-informed gradient-boosted surrogate (XGBoost/LightGBM) that trains in under 5s on CPU and delivers sub-millisecond inference; a K-shot calibration scheme cuts power error from 24.5% to 3.3% and wavelength error from 56.6% to under 1% on out-of-distribution macros.
- Integrated the surrogate with an NSGA-II optimizer to evaluate 100,000 CTS configurations in ~9s, with Pareto-optimal configs validated closed-loop in OpenROAD (power within 0.5%, skew within 5 ps).
- Led experimental design (LODO + OOD protocols), ablation studies, and manuscript preparation for both projects.

Machine Learning Undergraduate Researcher AUG. 2025 – PRESENT
Cyber Innovation Lab, USM ADVISOR: Dr. Nick Rahimi

- Proposed and developed a hybrid two-stage “filter-then-verify” framework using inductive Graph Neural Networks (GraphSAGE) and ModernBERT to detect multi-stage social engineering attacks.
- Constructed a three-phase structural anomaly detection pipeline (volume, relational, and contextual) to identify anomalous sender–receiver patterns in dynamic email networks.
- Achieved 92% precision by integrating a co-attention ModernBERT model to verify the semantic intent of flagged communications, reducing false positives on the Enron dataset.

Research Software Engineer NOV. 2025 – PRESENT
Institute for Advanced Analytics and Security, USM

- Active contributor to AQUAVIEW, a Python library exposing 7+ API endpoints for streamlined access to oceanographic data for NOAA, downloaded over 1,000 times.
- Processed 17M+ records spanning 125 years of the World Ocean Database, generated metadata in MongoDB, indexed in the cloud, and integrated into the UI via Elasticsearch.
- Implemented an automated LLM-based tagging system to clean metadata, standardizing Platform IDs for Uncrewed Marine Systems across 20M+ data points in 10+ sources.

INDUSTRY
EXPERIENCE

Software Engineering Intern (Backend) JUN. 2025 – AUG. 2025
Sports Media Inc (Dakdan Worldwide) Dover, DE

- Built Spring Boot microservices and REST APIs to support real-time ad-bidding workflows.
- Integrated Apache Kafka to power live inventory updates across 100+ venues.
- Contributed to CI/CD pipelines using Jenkins, Maven, and Git; deployed services on AWS and GCP, and wrote JUnit test suites in a Unix/Linux environment.

Web/App Developer Intern (Frontend) MAY 2024 – AUG. 2024
Crystal ERP Koshi, Nepal

- Migrated 4+ school websites from vanilla JavaScript to React, improving load performance by 25%.

- Introduced React + Tailwind UI components, reducing design code duplication by 20%.
- Streamlined UX consistency and reduced help-desk tickets by 33% through Redux-based state management and end-user training.

SPECIAL PROJECTS **Vinaya Journal** 2024 – PRESENT
Spring Boot, FastAPI, React, SQLite, Chroma, Electron.js [Live](#) · [GitHub](#)

- Built a privacy-first, full-stack desktop journaling app integrating a local LLM + RAG pipeline for offline AI journaling.
- Implemented RESTful Spring Boot endpoints with embedded SQLite, plus a FastAPI microservice with Chroma for semantic/vector search, LLM interactions via Ollama, and mood classification.
- 35+ active users; 150+ GitHub stars; available for Windows and Linux.

WordBuddy.ai 2024 – PRESENT
[GitHub](#)

- Built an interactive voice-driven learning platform for children using the OpenAI and ElevenLabs APIs.
- Delivered immersive pronunciation practice and engaging learning experiences.

HONORS & SCHOLARSHIPS **Academic Excellence Award — Full Tuition (4 years)** AUG. 2024
Freshman Designation Valedictorian Scholarship AUG. 2024
Honors Scholar Recognition AUG. 2024
1st Runner-Up, VOXO Hackathon NOV. 2024
Reclaiming Time with Generative AI
President’s List SPRING 2025

TALKS & TUTORIALS **Positive Uses, Negative Uses, and Limitations of Virtual Reality** 2024
MAS 89th Annual Conference, Biloxi, MS

- Presented findings on VR technology and VRET applications.
- Collaborated with peers Anish Kharal and Nhoojah Maharjan in CSC 309; supported by Dr. Sengupta.

TECHNICAL SKILLS PROGRAMMING: Python, Java, JavaScript/TypeScript, SQL, C++
ML / RESEARCH: PyTorch, HuggingFace Transformers, PEFT, WeightWatcher, LM-Eval-Harness, NumPy, pandas, Chroma
FRAMEWORKS: React, Spring Boot, FastAPI, Maven, Tailwind CSS, PostgreSQL
TOOLS & DEVOPS: Git, GitHub, Docker, AWS, GCP, Apache Kafka, Jenkins, JUnit, Postman
LANGUAGES: English (fluent), Nepali (native), Hindi (advanced)

RESEARCH INTERESTS Reinforcement Learning, Mechanistic interpretability, Physical Design, World Models, AI for Physical Design , ML in Cybersecurity